

Clave 10F07 e(D).

1) $h_F = 381 \text{ mts.}$ $t = 25 \text{ min} \rightarrow 1500 \text{ segundos}$

$h_m = 190.5 \text{ mts.}$

$M = 50 \text{ Kg.}$

a) $E_{p_o} = mgh_m$

$E_{p_o} = 50 \text{ Kg} \cdot 9.8 \text{ m/s}^2 \cdot 190.5 \text{ m}$

$\{ E_{Pot} = 93345 \text{ Joules.} \}$

b) $E_{p_F} = mgh_F$

$E_{p_F} = 50 \text{ Kg} \cdot 9.8 \text{ m/s}^2 \cdot 381 \text{ m}$

$\{ E_{Pot} = 186690 \text{ Joules} \}$

c) $W = mg(h_F - h_o)$

$W = 50 \text{ Kg} \cdot 9.8 \text{ m/s}^2 \cdot 190.5 \text{ m}$

$\{ W = 93345 \text{ Joules} \}$

$$d) P = \frac{W}{t}$$

$$P = \frac{93345 \text{ J}}{1500 \text{ s}}$$

$$P = 62.23 \text{ J/s}$$

$$\{ P = 62.23 \text{ Watts} \}$$

$$2) M = 1250 \text{ Kg}$$

$$V_0 = 75 \text{ Km/h} \rightarrow 20.8 \text{ m/s}$$

$$V_f = 45 \text{ Km/h} \rightarrow 12.5 \text{ m/s}$$

$$a) E_{K_0} = \frac{1}{2} m V_0^2$$

$$E_{K_0} = \frac{1}{2} (1250) (20.8 \text{ m/s})^2$$

$$\{ E_{K_0} = 270400 \text{ Joules} \}$$

$$E_{K_f} = \frac{1}{2} (1250 \text{ Kg}) (12.5)^2$$

$$\{ E_{K_f} = 97656.25 \text{ Joules} \}$$

$$b) W = \frac{m (V_f^2 - V_0^2)}{2}$$

$$W = \frac{1250 \text{ Kg} ((12.5 \text{ m/s})^2 - (20.8 \text{ m/s})^2)}{2}$$

$$W = -172743.75 \text{ Joules}$$

$$3) \text{ Potencia} = 90 \text{ Hp} \rightarrow 66186 \text{ Watts.}$$

$$\text{tiempo} = 6 \text{ h} \rightarrow 1296000 \text{ segundos.}$$

$$P = \frac{W}{t}$$

$$W = P \cdot t$$

$$W = 66186 \text{ Watts} \times 1296000 \text{ s.} \quad \frac{\text{Kg} \cdot \text{m}^2}{\text{s}^2} \cdot \text{s} \quad \text{Kg} \cdot \frac{\text{m}^2}{\text{s}^2} \cdot \text{s} \quad \text{J.}$$

$$W = 8.57 \times 10^{10} \text{ Joules}$$

4) a) Definir y dar 2 ejemplos

b) Conservativas : Peso, F. eléctrica, Elástica

No conservativas : F. rozamiento, Amortiguamiento
o disipativas

5) a) F

b) V